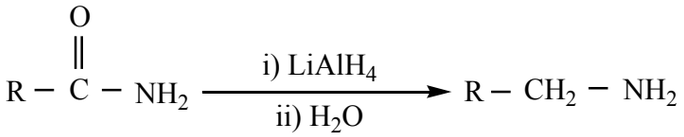
3) From amides

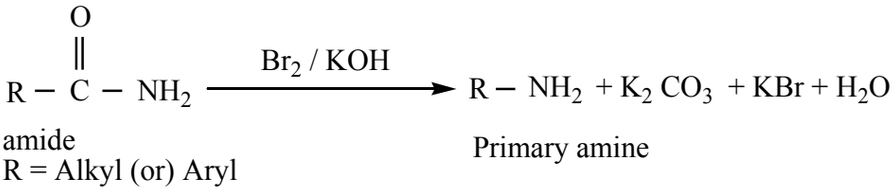
a) Reduction of amides with $LiAlH_4$ gives amines



b) Hoffmann's degradation reaction

When Amides are treated with bromine in the presence of aqueous or ethanolic solution of KOH, primary amines with one carbon atom less than the parent amides are obtained.

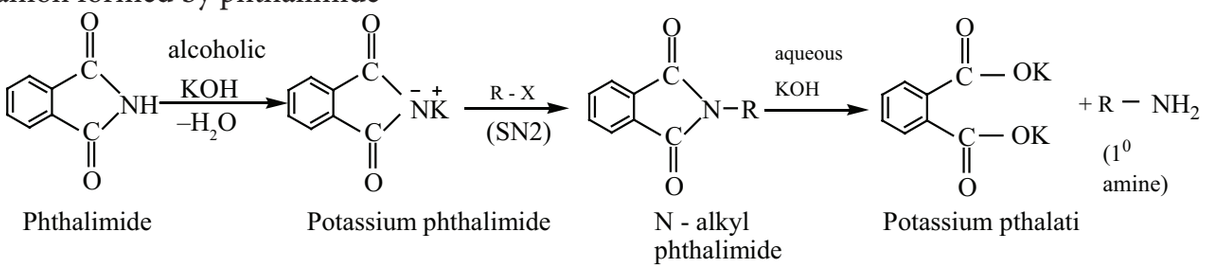
Example:



4) From alkyl halides

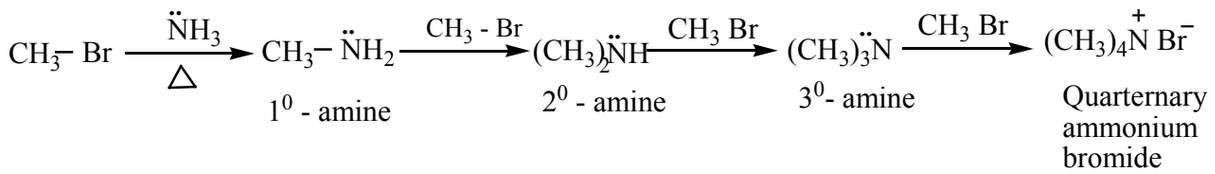
a) Gabriel phthalimide synthesis

Gabriel synthesis is used for the preparation of Aliphatic primary amines. Phthalimide on treatment with ethanolic KOH forms potassium salt of phthalimide which on heating with alkyl halide followed by alkaline hydrolysis gives primary amine. Aniline cannot be prepared by this method because the arylhalides do not undergo nucleophilic substitution with the anion formed by phthalimide



b) Hoffmann's ammonolysis

When Alkyl halides (or) benzylhalides are heated with alcoholic ammonia in a sealed tube, mixtures of 1^0, 2^0 and 3^0 amines and quaternary ammonium salts are obtained.



This is a nucleophilic substitution , the halide ion of alkyl halide is substituted by the $-NH_2$ group. The product primary amine so formed can also has a tendency to act as a nucleophile and hence if excess alkyl halide is taken, further nucleophilic substitution takes place leading

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